

Generated Volume Garden Workshop

Name: _____ Date: _____ Class: _____

Build, observe, calculate, and explain.

Geometry World uses the familiar building and exploration of block-based 3D sandbox games to make geometric measurement visible. Count cubes in a layer, connect layers to multiplication, and compare strategies. Explaining and checking those calculations provides practice toward accurate, flexible calculation fluency.

Compare the packing designs at two stations.

Lesson goals

1. Explain volume using equal layers.
2. Use multiplication and addition to verify a design.

Your lesson route

1. Packing model · 3 question steps

Print in portrait at 100% or Save as PDF. This companion follows the current lesson's tasks and question choices. Answer by hand; a drawing, words, or another clear representation can show your thinking. Use the world to gather observations before completing the paper independently.

Packing model

Guide: **Packing mentor** at world coordinates (0, 1, 1). Teaching models: packing-model. Use this sheet alongside the lesson world. Observe the named models or guide before answering; sketch and label the model so your reasoning can be understood on paper.

Construction / investigation task

Use the model to compare two layers with a one-layer arrangement.

Self-review: Check the task requirements, label units, and explain how your evidence supports your result.

Plan / sketch · label dimensions and units

Measure / calculate · record the quantities and the operations you choose

Explain and review

How does your multiplication represent the layers?

My mathematical reasoning and revisions

Packing model · Questions 1–2

Guide: **Packing mentor** at world coordinates (0, 1, 1). Teaching models: packing-model. Use this sheet alongside the lesson world. Observe the named models or guide before answering; sketch and label the model so your reasoning can be understood on paper.

Question 1 · Step 1

How many unit cubes fill the packing model?

A. 12 B. 6 C. 10

Sketch / label the model, or describe what you observe

Calculations · show an equation, units, and intermediate steps

Reasoning · explain your strategy and check your answer

My answer (include units): _____

Question 2 · Step 2

How many cubes are in each layer?

A. 3 B. 6 C. 12

Sketch / label the model, or describe what you observe

Calculations · show an equation, units, and intermediate steps

Reasoning · explain your strategy and check your answer

My answer (include units): _____

Packing model · Questions 3

Guide: **Packing mentor** at world coordinates (0, 1, 1). Teaching models: packing-model. Use this sheet alongside the lesson world. Observe the named models or guide before answering; sketch and label the model so your reasoning can be understood on paper.

Question 3 · Step 3

Explain a different arrangement with the same volume.

A. 2 by 2 by 3 B. 2 by 2 by 2 C. 3 by 3 by 3

Sketch / label the model, or describe what you observe

Calculations · show an equation, units, and intermediate steps

Reasoning · explain your strategy and check your answer

My answer (include units): _____

Connect the ideas

Choose one model or problem from this lesson. Explain how its representation connects to your calculations.

Model / diagram · show cubes, layers, dimensions, or a decomposition that fits the task

First calculation strategy · show the steps and units

Check with another strategy · explain why it agrees or what you revised

Reflection · what became easier and what do you still want to investigate?